WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER PROGRAM

AZNS12 ST-1 (NEAR #75), GADVAN WELL

Drilling Department

MARCH 2025

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45°c in summer, and Highest Temperature is 52°c. The Lowest Temperature is -4.4^{o}c in winter. Maximum Rainfall is 78 mm in February, Winter Season.

This Document covers the work over of well AZNS-122. Main headlines of program are de-complete the well and run new completion string.

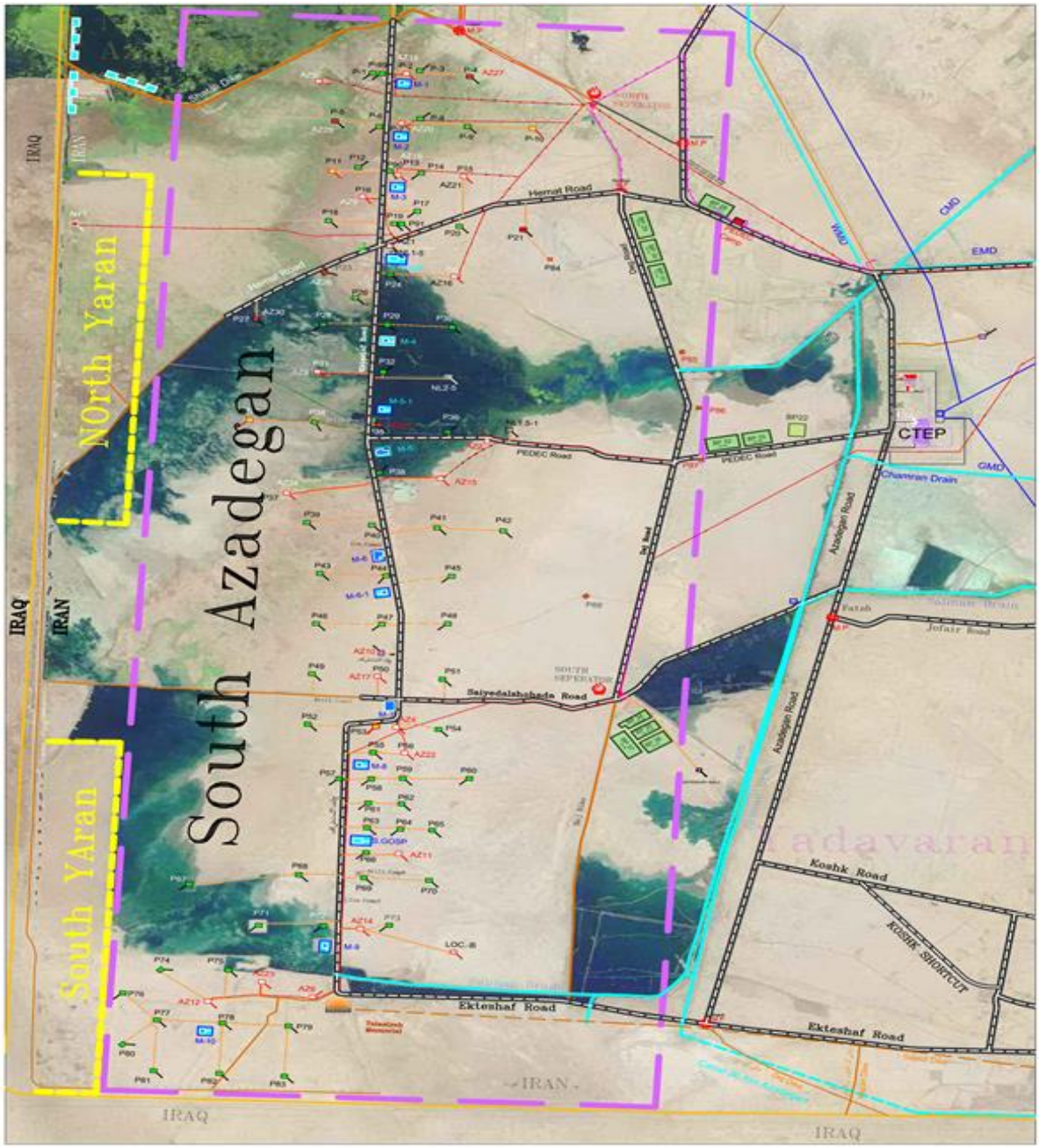


Figure 1: AZNS Oilfield Location Map

1.2 Well Information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Rig Name:	OEOC208
Type Of Operation:	Workover
Well Name / Type :	AZNS12 ST1 / Gadvan well in Block III
Pad Number:	Near75
Well Type:	Side Track
Wellhead Coordinate:	N: 189014.48 E: 3437769.78
TD Coordinate:	N: 189019.38 E: 3437748.34
Well Objective:	De-complete the well, Re-complete the well, Due to abnormal annulus pressure in 9-5/8" CSG X 4-1/2" TBG & left fish in hole
RTE:	14.45 m
GLE:	3.95 m
Well Duration :	91.9 days (Main Plan) + 136.2 days (Contingency Plan)
Total Depth (MD):	3945 m
Producer Formation:	Perforated in 7" liner (Exiting Perforation: 3867-3908 m)
Formation TD :	Gadvan
Reservoir :	Gadvan

Table 1: AZNS-12 information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.
Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 2: Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.
Block II is located between well AZN-010 and AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal
Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

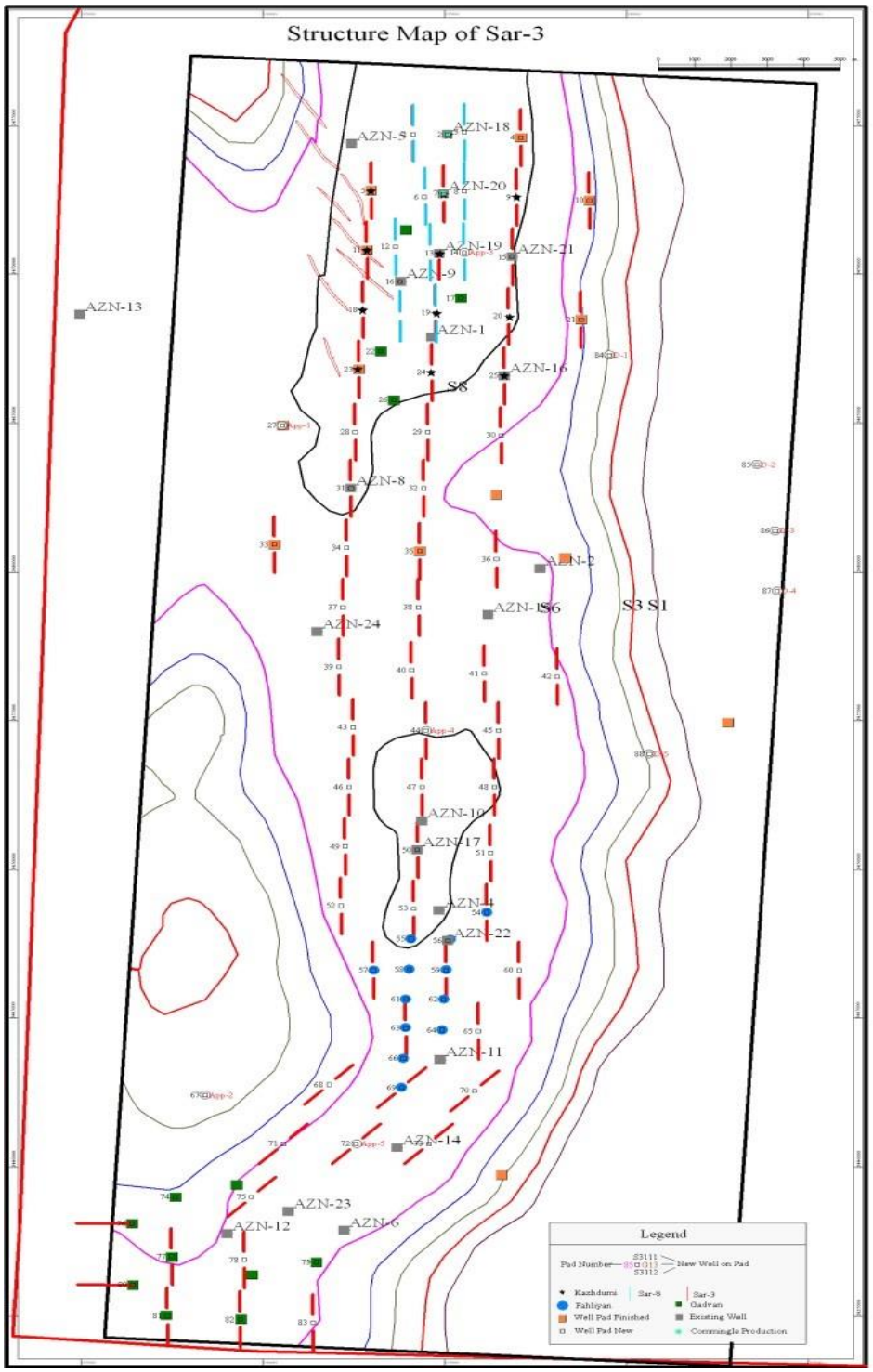


Figure 2: AZNS Area Distribution

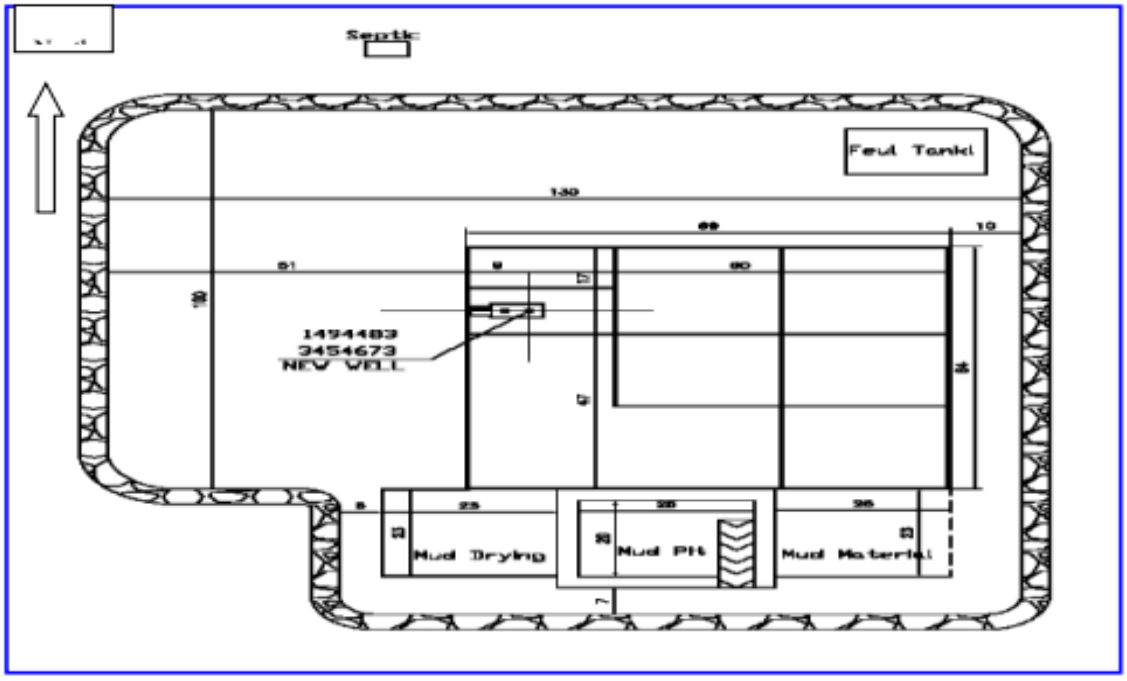
1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

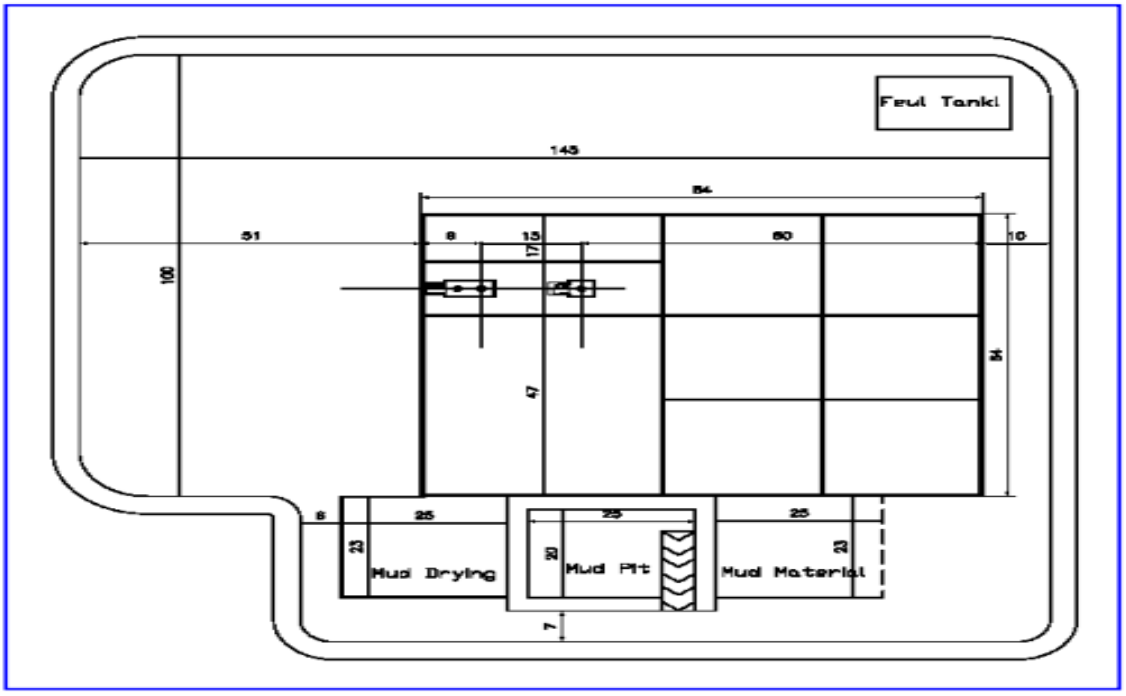
Table 3: Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm2
• Fuel tank concrete pond dimension: Length 20m×Width 12m×Height 0.5m
• Drainage: Width 30cm × Depth 30cm
• Burning pit: 15m×15m square, located on South-East or South-West with at least 350m distance from well head
• Cellar to cellar distance in multi-well well pad: 15m
• Conductor pipe depth under cellar bottom is 5.5m. The conductor pipe will be driven while constructing the well pad.
• Mud material dimension is 25m*35m +granular surface
• Mud drying dimension is 25*35m + concrete surface
• Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

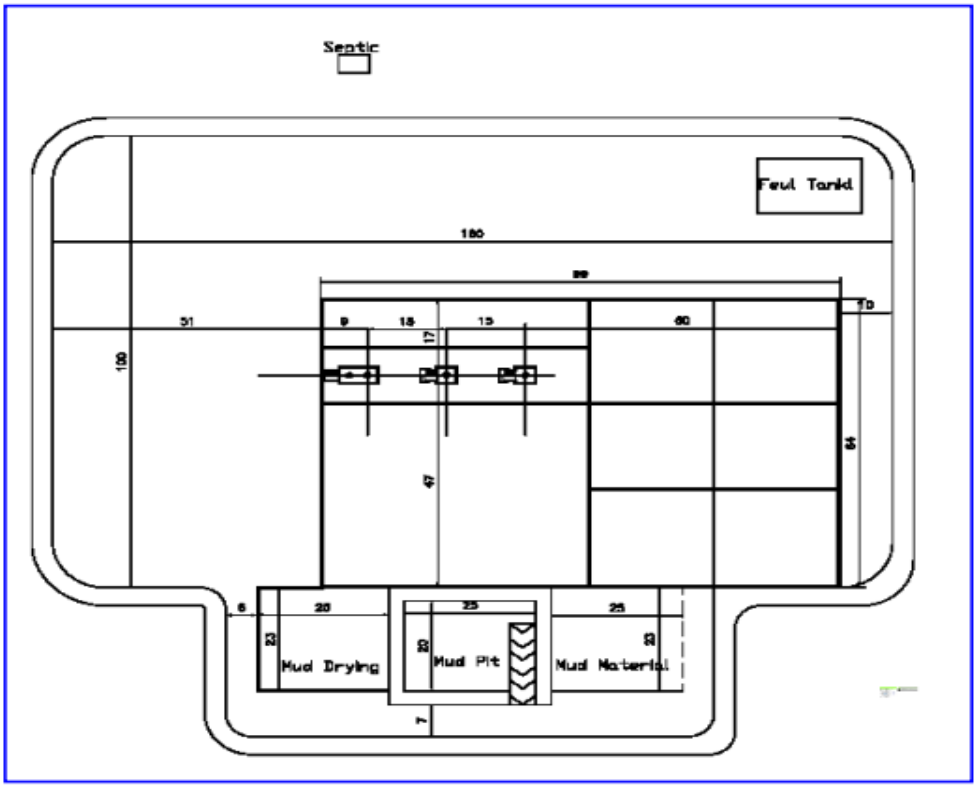
1. Types of well pads:
• One well in one pad.
• Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
• Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.



Well Pad for one well



Double-well Well Pad



Triple-well Well Pad

Figure 3: Well Pads Sketch

1.5 Reservoir Data

Table 4: Reservoir Data

Reservoir Name:	Gadvan	Unit
Fluids:	Oil	….
Datum Depth:	3850	M,TVDss
Datum Pressure:	Gad5a=4750 Gad5c=4750	Psi
Pressure Gradient:	0.3	Psi/ft
Temperature:	251	C°
Temperature Gradient:	1.32 F/100ft	1.32 F/100ft 0.73 C/100ft 2.4 C/100m
H2S:	0	PPM
CO2:	5300	PPM
Salt Salinity :	171000	ppm
Calcium Content:	9400	(mg/lit)

Reservoir	Sub-layer	Mid of Sub-LayerDepth in AZNS-12 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure
GADVAN	Gad-5a	-3690.7	244.3	1.32	4864	Dynamic Model
	Gad-5c	-3703.7	244.9		4874	Dynamic Model

Table 5: Predicted Pressure and Temperature and TTZ Contact

1.6 Formation Pressure Prediction in Block II

Predicted Formation Pressure in Block II				
Formation	**Interval**	**Pore Pressure**	**Collapse Pressure (Break-Out Pressure)**	**Fracture Pressure (Break-Down Pressure)**
	m	*SG*	*SG*	*SG*
Aghajari	0-974	0.98-1.18	0.06-1.39	1.01-1.89
Gachsaran	974-1281	1.13-1.31	0.78-1.36	1.89-2.01
Asmari	1280-1474	1.14-1.26	0.6-1.28	1.43-1.79
Pabdeh	1474-2124	1.20-1.26	0.82-1.36	1.40-1.78
Gurpi	2124-2539	1.21-1.26	0.66-1.02	1.40-1.76
Ilam	2539-2613	1.18-1.24	0.20-0.9	1.41-1.72
Laffan	2613-2620	1.13-1.18	0.24-1.11	1.41-1.72
Sarvak	2620-3261	1.14-1.26	0.24-1.23	1.44-1.81
Kazhdumi	3261-3465	1.1-1.14	0.46-0.96	1.93-1.94
Dariyan	3465-3627	1.18-1.25	0.55-1.1	1.96-2.06
Gadvan	3627-3932	1.1-1.25	0.55-1.44	1.97-2.11

Table 6: Formation Pressure Prediction

1.7 Drilling History

26" Hole Section

Spud the rig FATH 64 on 1387.04.15 at 12:00 o'clock.

Commenced drilling 26" hole section from 14m to 155.5m. Run 20" casing (J-55m 133#, BTC) to 154m, perform cemented 20" casing (S.W:115 pcf), got CMT in return. Install 20" 3K C.H.H (type ABB).

17 1/2" Hole section

Drilling of 17-1/2" hole section started on 1387/05/20 from 154m to 1259m with No Loss- No Gain. Perform condition trip. Run CDR/GR log.

R/U 13-3/8" casing running equipment, held PJSM and run 13-3/8" casing with 13-3/8" float shoe (Nasr Mobtakeran) + 2 Jt 13 3/8" CSG + 13-3/8" float Collar (Nasr Mobtakeran) + 106 Jts 13 3/8" CSG (C-95, 72#, BTC to 1259 m. perform Cement 13-3/8" casing (883 bbl head CMT, S.W: 90 pcf, 135 bbl tail CMT S.W:115 pcf). Install DSA 20" 3 K X 16 ¾" 5k & 16 ¾" 5K (MCEVOY) X 13 5/8" 5K (ABB).

12 1/4" Hole section

Drilling of 12-1/4" hole section started on 1387/06/10 from 1259 m to 1585 m, Obs. complete loss, cure the loss by pumping LCM, Cont. Drlg to 1600 m. Obs. No return. POOH to surface, RIH with C.Cut bit, spot 120 bbl 94 pcf CMT, POOH to surface, RIH with bit, Cont. Drlg formation to 1860 m with heavy loss (cure the loss using LCM pill), @ 1860 m, Obs. 500 psi pressure drop, POOH to surface, Obs. 12 ¼" bit + bit sub + 2 Jts 8 ½" D. Collar left in hole. Retrieve the fish, RIH with open end drill pipes, pump 150 bbl CMT 94 pcf in 2 stage, D/O CMT, Cont. Drlg to 2130 m, 50-80 BPHL. stuck string, work on stuck, free the pipe, POOH to surface, obs. String twist off from Jar mandrel. Retrieve the fish. riH W/ BIT & Cont. Drlg formation to 2245 m, Obs. Pressure drop, Obs. 92.26 m fish left in hole. Retrieve the fish. RIH with bit & Cont. Drlg formation to 3792 m (section TD). Perform condition Trip, run BHC/HRLA/TLD/HGNS/GPIT/HNGS log, POOH to surface, run MDT log. Perform 230 bbl balance CMT plug in 2 stage. RIH with D/O BHA, D/O CMT & to 3450 m (9 5/8" CSG point), POOH to surface.

R/U CSG running equipment, held PJSM and run 9 5/8" casing with 9 5/8" float shoe (IDC) + 2 Jt 9 5/8" CSG (L-80, 53.5#, NEW VAM) + 9 5/8" float Collar (IDC) + 98 Jts 9 5/8" CSG (L-80, 53.5#, NEW VAM) + 185 Jts 9 5/8" CSG (L-80, 47#, VAM) + 2 Jts 9 5/8" CSG (L-80, 53.5#, VAM) to 3448 m. drop ball & apply pressure to 4500 psi & try to Circ. No Success. Set CSG slips, install DSA 12" 5K X 3K, T.H.S 12" 3K 10" 5k (Toloo Khoozestan), RIH with ounching tools & punch 9 5/8" CSG F/ 3419 to 3420 m. try to perform Circ. No success.

RIH with punching tools & punch 9 5/8" CSG F/ 3408.5 to 3410.5 m. perform Circ., OK. RIH with RTTS PKR, perform CMT job, unset PKR & POOH to surface.

8 1/2" Hole section

RIH with bit, D/O CMT to 3468 m (KOP). POOH to surface. RIH with Dir. Drlg BHA to 3468 m. start Dir. Drlg F/3468 to 3735 m. left fish in hole, retrieve fish, RIH with bit & unable to pass 3660m in several times. Displace the well with O.B.M, Cont. Drlg to 3846 m. perform condition trip, POOH to surface.

R/U 7" LNR running equipment. M/U 7" F. Shoe + 1 Jt 7" LNR + 7" F. Collar + 2 Jt 7" LNR + 7" LDC + 19 Jts 7" LNR (C-75, 29 #, N.VAM) & RIH to 3846m, perform CMT job.

6 1/8" Hole section

RIH with Dir. Drlg BHA, Drlg formation to 3945m (section TD), perform condition trip. Perform logging operation (AIT/BHC/LDT/CNT/GR)

R/U LNR running equipment, M/U 5" F.Shoe + 29.5 5" Blank LNR (L-80, 18#, HTS) + 47.8 m 5" LNR perforated + X.O.S + 1 Jt 5" LNR 18# VFJ + LDC + 6 Jts VFJ LNR + 5" LNR HGR & RIH to 3945 m. set HGR, POOH to surface, RIH with 4 1/8" E. Mill & clean out BHA to 3945 m. displace the well with Cacl2, POOH to surface.

Completion

Run completion string with 9 Jts 4 ½" TBG + 291 Jts 7" (L-80,29#, N.Vam TBG + 3SV + 5 No. fluted adapter + 5 Jts 7" TBG + fluted swage + TBG HGR (10" X 7" X 9 5/8" neck) set PKR @ 3334m.

1.8 Well Montage

Not Available.

1.9 Casing & Liner Depth

Casing Spec.	20"	J-55, 133#, BTC	0-154 m
	13-3/8"	C-95, 72#, BTC	0-1259 m
	9-5/8"	L-80,47 & 53#, VAM & N. VAM	0-3448 m
	7"	C-75, 29#, N. VAM	3285 – 3846 m
	5”	L-80, 18#, VFJ, HTS	3779-3945

Table 7: Casing & Liner Data

1.10 Well Schematic before Workover

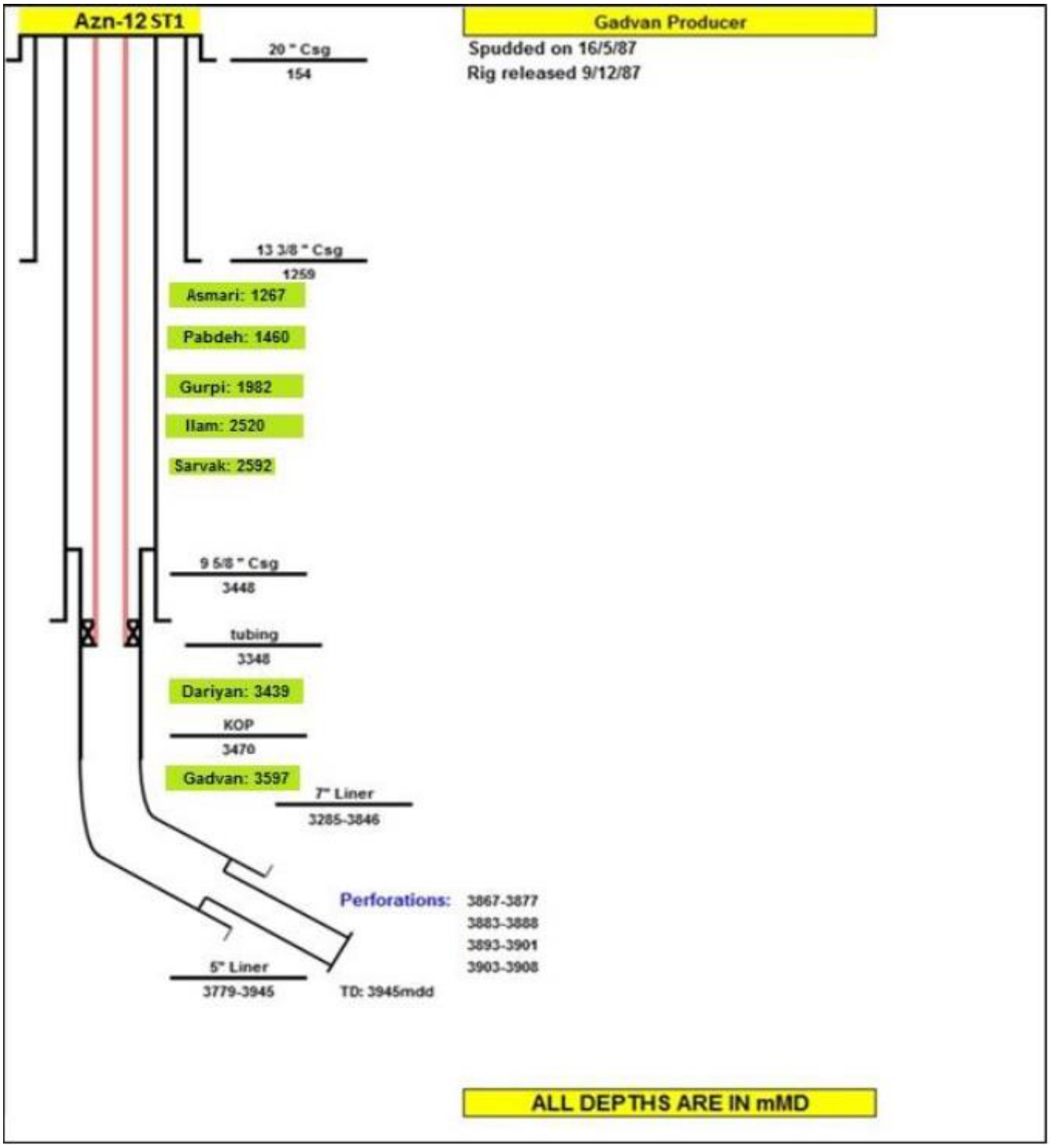
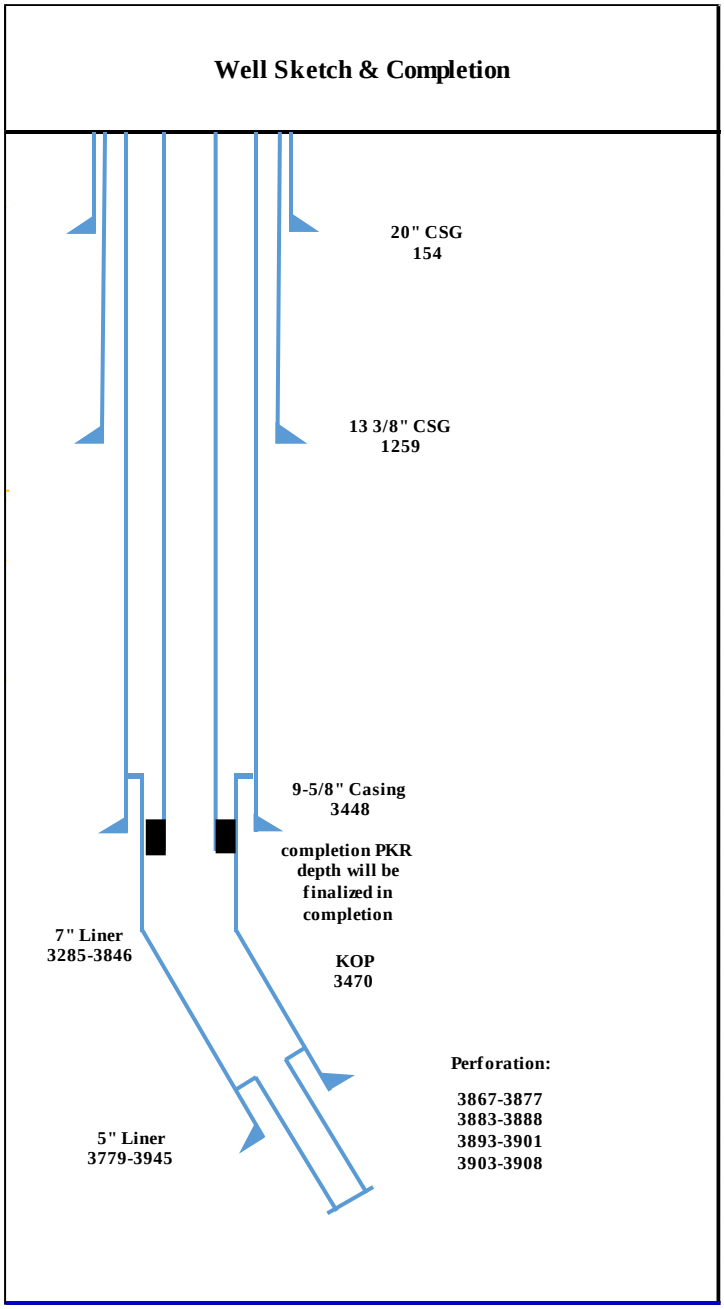


Figure 4: Well Schematic Before workover

1.11 Well Schematic after Workover

Figure 5: Well Schematic After workover



2. Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
Miocene	Fars	Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi Tarbur L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
		Laffan		Claystone	2691	2642	2625
Turonian Cenomanian	Bangestan	Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian	Bangestan	Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian	Khami	Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian	Khami	Fahliyan		Limestone, Marl	4092	3989	3974

Figure 6: Azadegan Stratigraphic Column

3. Logging Program

Hole Size	Casing Size	Logs
		Cased Hole
12 ¼”	9 5/8”	Corrosion Log
8-1/2"	7" Liner.	SBT/VDL/GR/CCL Corrosion Log

Table 8: Logging Program

4. Program Outline

4.1 Contractors Involved

Table 9: Contractor Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 201
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	GEO DATA
6	H2S Monitoring and Control	Pars Hafaran Arvand BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Well Head	OEOC

4.2 Time Breakdown

Work-Over Operation TBD of Well: AZNS-12 Work-over (Gadvan Producer)	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over												
Held PJSM. Spud the Rig. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC.	Spud		Ga.	3945		0	0.00	0.0	0.0	0.0	0.00	0.0
Check the wellhead for any damage or leakage. Record well head pressures; check all annulus pressure, Bleed off any annulus pressure to Zero.	Wellhead		Ga.	3945		24	1.00	1.0	1.1	26.4	1.10	1.1
Close XMT valves. Remove 7 1/16" XMT cap. R/U Well Test lines from XMT to burning pit. And connect f/ 7"-9 5/8" annulusside out let valv e to Pump Truck/mud pumps.	Well Test		Ga.	3945		12	0.50	1.5	0.6	13.2	0.55	1.7
Release well pressure through Burning Line in Safe Mode to Burning Pit & Also Monitor W.H.Pressure.	Well Test		Ga.	3945		12	0.50	2.0	0.6	13.2	0.55	2.2
R/UP Slick Line & Test Lubricator,Retrieve sub surface safety valve, install dummy safety valve. Note:Suitable Dummy Safety Valve Shall be Provided by Client. Note:if Retreiving 3SV is not Possible follow contingency plan#01	De-Completion		Ga.	3945		12	0.50	2.5	0.6	13.2	0.55	2.8
R/U SL, RIH w/ Gauge Cutter To TOF @ 2800 m.	De-Completion		Ga.	3945		12	0.50	3.0	0.6	13.2	0.55	3.3
R/U CT. RIH and tag top of the fish @ 2800m.Note: Chemical wash may be needed while RIH with CT .	De-Completion		Ga.	3945		36	1.50	4.5	1.7	39.6	1.65	5.0
Displace the well with heavy mud to make the well overbalanced.Observe the annulus, POOH. R/D CT.Note: If wellhead pressure is not zero, follow contingency plan #2.	De-Completion		Ga.	3945		18	0.75	5.3	0.8	19.8	0.83	5.8
Install TWCV in Tubing Hanger. Note:Suitable BPV Shall be Provided by Client.Note:if Unable to Set BPV Due to Damaged Profile,Follow Contingency Plan#3	De-Completion		Ga.	3945		12	0.50	5.8	0.6	13.2	0.55	6.3
N/D X.M.T, N/U BOP and perform BOP pressure test.retrieve TWCV	De-Completion		Ga.	3945		48	2.00	7.8	2.2	52.8	2.20	8.5
R/U & RIH W/Colliding Tools & Collide Completion String from 2800m ,POOH & L/D, displace and overbalance the well to get in & out uniform MW.Observe the annulus,	De-Completion		Ga.	3945		168	7.00	14.8	7.7	184.8	7.70	16.2
RIH with TBG hanger retrieving tool and engage with TBG hanger. Note:if Tubing Hanger Thread damaged follow Contingency Plan#04	De-Completion		Ga.	3945		12	0.50	15.3	0.6	13.2	0.55	16.8
Unlatch, Displace completion fluid & oil in hole with proper kill mud and pull completion string recautionary due to probable trapped pressure below fish. While POOH monitor the wells and annulus to make sure the well is safe and over balance. (from 3333.85m). Note:if unable to unlatch completion string due to string pack off with huge amount of debries follow contingency plan#05 Note:if while trying to unlatch completion, string was part due to string wash out follow contingency plan#06	De-Completion		Ga.	3945		160	6.67	21.9	7.3	176.0	7.33	24.1
RIH with proper size Spear to TOF, Try to catch & Retrieve fish in several attempts then POOH.	Trip		Ga.	3945		120	5.00	26.9	5.5	132.0	5.50	29.6

RIH with 8 ½" Mill and clean out BHA (mill remaining fish if Not Retrieved in previus Step) to 7" L.LAP (3285m), CBU, POOH.	Trip		Ga.	3945		310	12.92	39.8	14.2	341.0	14.21	43.8
RIH with Mill to 3333.85m (P/U required DP jt by jt), Mill permanent packer, chase the packer to Top of 5" Liner Lap @ 3779m (or where ever PKR stopped), CBU, POOH.	Trip		Ga.	3945		192	8.00	47.8	8.8	211.2	8.80	52.6
RIH with Taper Tap and catch the packer, recover completion packer.POOH To Surface. Note:if string stuck while Recovering Completion Packer follow contingency plan#07	Trip		Ga.	3945		96	4.00	51.8	4.4	105.6	4.40	57.0
RIH with 6" or 5-7/8 bit and clean out BHA to 5" Liner lap (3779m), CBU shakers clean, POOH.	Trip		Ga.	3945		60	2.50	54.3	2.8	66.0	2.75	59.8
RIH with 4 1/8" Mill and clean out BHA to bottom (TD: 3945m), take care inside pre-perforation interval (3867m - 3908m), CBU till hole clean, POOH to surface.	Trip		Ga.	3945		60	2.50	56.8	2.8	66.0	2.75	62.5
M/U Tandem scraper BHA, RIH with tandem scrapers including 9-5/8" casing scraper & 7" liner scraper to top of 5" liner (3779m). Scrap 9-5/8" casing and 7" liner and L/D same.	Trip		Ga.	3945		72	3.00	59.8	3.3	79.2	3.30	65.8
RIH & Perform TLC Logging (SBT/CBL/VDL) Inside 7" Liner,POOH to Surafce.	Log		Ga.	3945		72	3.00	62.8	3.3	79.2	3.30	69.1
RIH & Perform TLC Logging (Corrosion Logging) Inside 7" Liner,POOH to Surafce.	Log		Ga.	3945		72	3.00	65.8	3.3	79.2	3.30	72.4
RIH & Perform WL Logging (Corrosion Logging) Inside 9-5/8" Liner,POOH to Surafce. Note: for Change THS follow contingency plan#8	Log		Ga.	3945		41	1.71	67.5	1.9	45.1	1.88	74.3
RIH and set RTTS in 9 5/8" CSG (3270m) as a barrier.	Wellhead		Ga.	3945		0	0.00	67.5	0.0	0.0	0.00	74.3
N/D BOP, Change DSA pack off to new one. Perform pressure test P-seals. N/U BOP and pressure test.Note:DSA pack off Shall be Provided by Client.	Wellhead		Ga.	3945		0	0.00	67.5	0.0	0.0	0.00	74.3
RIH with DP and retrieve RTTS, POOH to surface.	Wellhead		Ga.	3945		0	0.00	67.5	0.0	0.0	0.00	74.3
RIH w/ DP's and DC's, POOH & L/D same.	Trip		Ga.	3945		0	0.00	67.5	0.0	0.0	0.00	74.3
RIH scraper and scrape 7" production packer setting area, displace the well with Calcium Chloride (CaCl2) as completion fluid, POOH to surface.	Trip		Ga.	3945		68	2.83	70.4	3.1	74.8	3.12	77.4
Run 7.5K completion string as per program.	Completion		Ga.	3945		168	7.00	77.4	7.7	184.8	7.70	85.1
Set BPV inside TBH, N/D BOP's & install new XMT.	Completion		Ga.	3945		58	2.42	79.8	2.7	63.8	2.66	87.8
Retrive BPV, install TWCV and Pressure test the XMT.	Completion		Ga.	3945		18	0.75	80.5	0.8	19.8	0.83	88.6
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.	Completion		Ga.	3945		8	0.33	80.9	0.4	8.8	0.37	89.0
Retrieve TWCV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0

Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
RIH with S/L, retrieve protection sleeve.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion		Ga.	3945		Rig Less	0.00	80.9	0.0	0.0	0.00	89.0
Total Duration of 6-1/8" Hole Section:							***80.9***	***80.9***			***89.0***	***89.0***
NOTE:												
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip			3763		20	0.83	81.7	0.9	22.0	0.92	89.9
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service			3763		44	1.85	83.6	2.0	48.9	2.04	91.9
Total Duration of AZNS-12 Work-over Well:								**83.6**				**91.9**
Contingency Plans TBD of Well: ~~AZNS-12 Work-over~~	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Based on operation situation in each step, the following contingency plans might be followed												
Contingency Plan#01 R/U CT RIH & Wash 3SV by Chemical wash & Try to Retrive 3SV by CT if no Success Remedial Plan will be Prepared in advance by Operator & Will be Follow After Client approval.	De-Completion		Ga.	3945		480	20.00	20.00	22.0	528.0	22.00	22.0
Contingency Plan#02 R/U & RIH W/Coliding Tools & Colide Completion String from Proper Depth,POOH & L/D, Displace well with heavy mud and try to kill the well,Circulate & Balance MW.	De-Completion		Ga.	3945		360	15.00	35.00	16.5	396.0	16.50	38.5
Contingency Plan#3 Install 3SV Blanking plug or 7 inch Wire Line Bridge Plug	De-Completion		Ga.	3945		60	2.50	37.50	2.8	66.0	2.75	41.3
Contingency Plan#04 RIH w/Spear & Try to engage w/Tubing Hanger	De-Completion		Ga.	3945		48	2.00	39.50	2.2	52.8	2.20	43.5
Contingency Plan#05 R/U & RIH with FPIT tools to check string free point,POOH to Surface & L/D.R/U & RIH W/Coliding Tools & Colide Completion String from Proper Depth,POOH & L/D.POOH Completion String from Coliding Depths in Several Runs.	Completion		Ga.	3945		408	17.00	56.50	18.7	448.8	18.70	62.2
Contingency Plan#06 Pull Completion String untill string will be part from weaked point then POOH & L/D String in Several attempt. It is assumed to part from 4- 1/2" expansion joint as weak point. Mill the remainder of fish & packer. Follow the rest of main plan.	Completion		Ga.	3945		728	30.33	86.83	33.4	800.8	33.37	95.5
contingency plan#07 RIH w/5-7/8" or 6" Mill & Mill the Packer & Tail in Several Runs, POOH to Surafce & L/D mill.	Trip		Ga.	3945		288	12.00	98.83	13.2	316.8	13.20	108.7
contingency plan#08 RIH and set RTTS in 9 5/8" CSG as a barrier. N/D BOP, Change DSA pack off to new one. N/U BOP and pressure test.Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC. Note:DSA pack off Shall be Provided by Client.	Trip		Ga.	3945		600	25.00	123.83	27.5	660.0	27.50	136.2
Total Duration of Contingency Plan AZNS-12 Work-over Well:								**98.8**				**136.2**

5. Sequence of Operation

5.1 Main Objective

Decision has been made to de-complete he well & re-complete with new completion equipment. Check and smooth collapse interval from 3889-3911m.

5.2 Main plan

1. Held PJSM. Spud the Rig.
2. Check the wellhead for any damage or leakage. Record well head pressures; check all annulus pressure.
3. Apply pressure in control line & check same for any leakage, try to open 3SV (check indication of opening 3SV flapper) & monitor well pressure again.
4. Secure the well by closing all XMT valves except BTM master valve.
5. L/U pump truck to 7 1/16" 5K sweep bend & L/U F/ 7" to 9 5/8" completion annulus to rig choke manifold, release annulus pressure through burning pit & monitor pressure.
6. Remove 7 1/16" XMT cap, R/U 7 1/16" 5K lubricator & pressure test same against Deeping valve, R/U slick line meanwhile monitor completion annulus pressure.
7. RIH with gauge cutter to top of 3SV, POOH to surface. RIH with pulling tools of WLR 3SV (3SV Spec: Baker, Landing Nipple: 5.950", AF), Retrieve 3SV, POOH to surface, RIH with dummy valve, set dummy in 3SV profile, POOH to surface.

Note: If unable to RIH gauge cutter with slick line or unable to retrieve 3SV with slick line, follow contingency plan #1.

8. RIH with gauge ring to TOF (2800 m) , POOH to surface & L/D same. R/U CT & RIH with CT to TOF while pumping chemical wash, Displace inside completion string with proper mud weight (85 pcf)) POOH CT to surface & R/D same.

Note: if wellhead pressure is no zero, follow contingency Plan#2.

9. Install TWCV in TBG HGR, perform pressure test same, open tie down screw & pressure test same. (TWCV should be provided by client.)

Note: If unable to set TWCV in TBG HGR due to damaged TBG HGR profile, follow contingency Plan#3.

10. N/D X.M.T
11. N/U 13 5/8” 10 K BOP stack & pressure test same as per NIOC Policy.
12. Retrieve TWCV.
13. P/U & M/U L.J & TBG HGR running tools, screw on TBG HGR. (6 ½” fishing Jar should be ready in location by OEOC)

Note: In case of unable to screw L.J on TBG HGR follow contingency Plan#4.

14. Unlatch completion string from Anchor Seal (Baker, KC-22) (Approximately 291 Jts 7” TBG (L-80, 29#, VAM) between 3SV & X.O.S and 5 Jts above 3SV and 9 Jts 4 ½” TBG 13.5#, L-80, VAM between XOS & expansion Jt. (Approximate length of 7” TBG: 3250 m, Yield Strength of 7” TBG 29#, L-80 is 676 Klb, maximum make up TRQ of 7” TBG 29pp, L-80, VAM is 15250 ft.lb. Yield strength of 4 ½” TBG 13.5#, L-80, is 307 Klb. maximum make up TRQ of 4 1/2” TBG 13.5pp, L-80, VAM, is 6000 ft.lb)

Note: In case of unable to unlatch completion string from Anchor Seal, due to string pack off with huge amount of debris follow contingency Plan#5.

Note: In case of while trying to unlatch completion string, the string was parted from weak point due to string wash out, follow contingency Plan#6.

15. POOH completion string to surface precautionary due to probable trapped pressure below fish. While POOH monitor the wells and annulus to make sure the well is safe and over balance.
16. P/U & M/U & RIH with proper fishing tools to TOF, try to catch & retrieve fish in several attempts the POOH to surface.
17. P/U & M/U 8 3/8” M.M.J Mill & 5” DP as Jt by Jt & RIH with clean out BHA to 7” L. Lap (3285 m). confirm 7” L. Lap, Pump Hi-Vis pill to sweep the well, POOH to surface.
18. P/U & M/U 6” or 5 7/8” M.M Junk Mill & 63 Jt 3 ½” DP as Jt by Jt & RIH, mill completion PKR & chase same to 5” L.Lap (3779 m) while pump Hi-Vis periodically, CBU, POOH to surface.
19. RIH with 2”*4” Taper tap & catch tail of completion string, recover the PKR, POOH to surface.

Note: if string stuck while recovering completion PKR follow contingency plan#7

20. RIH with 6” or 6 1/8” Mill and clean out BHA to 5" Liner lap (3779m), pump Hi-Vis pill & CBU till shakers clean, POOH to surface.
21. RIH with 4 1/8” Mill 15 Jt 3 1/8” DC & 15 Jts 2 7/8” DP as Jt by Jt and clean out BHA to BTM (TD: 3945m), take care inside pre-perforation interval (3867m – 3908m), CBU till hole clean, POOH to surface.
22. M/U Tandem scraper BHA, RIH with tandem scrapers including 9-5/8" casing scraper & 7" liner scraper to top of 5" liner (3779m). Scrap inside 9-5/8” casing and 7” liner to 5” L.Lap, Pump Hi-Vis pill & CBU to sweep the well, POOH to surface and L/D same.
23. P/U & M/U logging equipment & Perform (SBT/CBL/VDL) log from Inside 7" Liner, as TLC method, POOH to Surface.
24. Perform Corrosion log from inside 7” LNR as TLC method. POOH to surface & R/D same.
25. R/U shooting nipple, M/U wireline equipment, take corrosion log from 9 5/8” CSG to surface. R/D wireline equipment.

    Note: for change THS Follow Contingency plan #08:

    Note: Based on corrosion log, tie back operation will be needed, contractor will perform tie-back operation as extra work if needed.
26. P/U & M/U 6” or 6 1/8” Tri Cone bit & 7” scraper & RIH with scraper BHA & scrap 7” completion PKR setting depth, pump Hi-Vis pill & sweep the well, displace the well with CaCl2 as completion fluid, POOH to surface & L/D all tubulars.
27. Run 7.5 K completion string as per program.
28. Set BPV inside TBH, N/D BOP's & install new 10K XMT.
29. Retrieve BPV, install TWCV and Pressure test the XMT.
30. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.
31. Retrieve TWCV, R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.
32. RIH with CT to bottom, displace the well with diesel or nitrogen & observe well, POOH.
33. R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.

34. Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.
35. R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L
36. RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve"
37. Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs.

    Note: The time is only for flowing duration (R/U, R/D & preparation time is excluded).

    Note: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.
38. RIH with S/L, set protection Sleeve, POOH & R/D SL.
39. RIH with C/T to bottom & displace the well with diesel, POOH.
40. RIH with S/L, install plug & prong into NO GO landing nipple, POOH.
41. RIH with S/L, retrieve protection sleeve.
42. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.

Contingency Plans of Well AZNS12 ST-1 Work-over:

Based on operation situation in each step, the following contingency plans might be followed:

Contingency plan #1: R/U CT & RIH & Wash 3SV by Chemical wash & Try to Retrieve 3SV by CT if no Success Remedial Plan will be Prepared in advance by Operator & Will be Follow After Client approval.

Contingency plan #2: R/U Wireline & PCE (pressure control equipment), RIH with punching tools, (colliding tools should be on rig site as back up) punch 4 ½" TBG from top fish, POOH WL to surface & R/D same. Perform direct Circ. To balance mud weight in & out. Observe the well.

Contingency plan #3: Install 3SV blanking plug or 7" WL bridge plug.

Contingency plan #4: RIH with proper spear & engage with TBG HGR.

Contingency plan #5: R/U & RIH with FPIT tools to check string free point, POOH to surface & L/D. R/U & RIH with colliding tools & collide completion string from proper depth, POOH to surface & L/D same. POOH completion string to surface precautionary due to probable trapped pressure below fish. While POOH monitor the wells and annulus to make sure the well is safe and over balance.

Contingency plan #6: pull completion string (maximum 80% of TBG yield strength) until string part from weak point then POOH & L/D string. it is assumed to part from 4 ½" expansion joint as weak point. (Approximately 291 Jts 7" TBG (L-80, 29#, VAM) between 3SV & X.O.S and 5 Jts above 3SV. (Approximate length of 7" TBG: 3250 m, Yield Strength of 7" TBG 29#, L-80: 676 Klb, Yield strength of 4 ½" TBG 13.5#, L-80 is 307 Klb). POOH completion string as main plan, RIH with proper Fishing tools, try to catch the fish & work on unlatch completion string from anchor seal, POOH rest of completion string to surface. If No Success, Mill the remainder of fish & packer. Follow the rest of main plan.

Contingency plan #7: RIH w/6" or 6 1/8" Mill & Mill the tail of completion string & Tail in Several Runs, POOH to Surface & L/D mill.

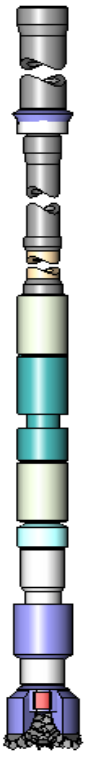
Contingency plan #8: RIH and set RTTS in 9 5/8" CSG as a barrier. N/D BOP, Change DSA pack off to new one. (or change section C) N/U BOP and pressure test.

Note: DSA pack off Shall be Provided by Client.

6. BHA Program

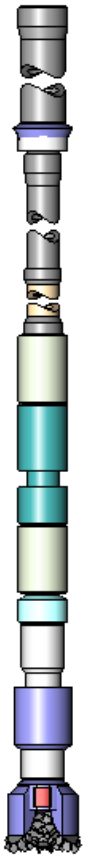
6.1 8-3/8" C/O BHA

BIT SIZE (inch)	8-3/8"	BHA Configuration to C/Out Inside 9-5/8" Casing				Collar	OD
Depth In/Out (m)						DC-1	6-1/2"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	85						
B.FACTOR	0.827						
Weight in Air (lb)	46,880						
Bouyed Weight (lb)	38,747						
Bouyed Weight Below Jar (lb)	29,532						

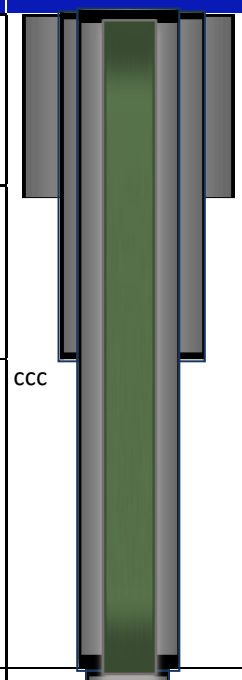
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" HWDP	3	5	3	4-1/2" IF Pin x Box	27.42	173.57
	6-1/2" Drill Collar	3	6-1/2	2-13/16	4-1/2" IF. P x B	27.42	146.15
	6-1/2" Jar	1	6-1/2	2.75	4-1/2" IF. P x B	8	118.73
	6-1/2" Drill Collar	12	6-1/2	2-13/16	4-1/2" IF. P x B	109.68	110.73
	6-1/2" Bit Sub W/ Float Valve	1	6-3/4"	3	4-1/2" Reg. B x IF B	0.75	1.05
	8-3/8" M.M.J Mill	1	8-1/2"	-	3-1/2" Reg. P x B	0.3	0.3

6.2 6 1/8" BHA

Mill SIZE (inch)	6 1/8" Mill	BHA Configuration to C/Out Inside 7" Liner	Collar	OD
Depth In/Out (m)			DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0.784			
Weight in Air (lb)	31,855			
Bouyed Weight (lb)	24,963			
Bouyed Weight Below Jar (lb)	17,907			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	-1/2" Heavy Weight Drill Pip	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. B x 3-1/2" IF B	0.75	1.05
	6 1/8" Mill	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

7. Mud & Cement Program

Hole Size (in)	Casing Size (in)	Casing Program	Depth (m MD)	Drilling Fluid system	Mud Properties
26”	20”		154 m	**Already Has been Drilled**	
17 1/2”	13 3/8”		1259 m	**Already Has been Drilled**	
12 1/4”	9 5/8”	ccc	3448 m	**Already Has been Drilled**	
8 1/2”	7’’ Liner		From 3285 m To 3846 m	**Already Has been Drilled**	
6 1/8”	5” Liner		From 3779 m To 3945 m	**Already Has been Drilled**	

Fluid Properties Summary		
MUD PARAMETERS		Interval
		I
Bit diameter	inch	
Density	pcf	72 - 85
Marsh Viscosity	sec	38 – 45
PV	cps	ALAP
Yield Point	lb/100ft²	12 – 18
Gel 10 Sec	lb/100ft²	3 - 5
Gel 10 min	lb/100ft²	4 - 8
API Filtrate	cc/30'	< 20
pH	-	9 – 10
Ca++	mg/l	< 1200
Total Hardness	Mg/l	<1600
MBT	ppb	< 5
LGS (drilled Solids)	% vol	
CL^-	Mg/l	120000-188000

Mud volume

DRILLING FLUID TYPE					
Mud Density (pcf)	72	- 85	LGS Below (%vol)	6	
Bit Size (inch)	0		Solid Control Equipment Eff. %	80	
Section TD (m)	3945		Hole Wash Out %	0	
Casing OD/ID (inch)	9.625 / 8.835		Dilution Factor (bbl/m)	0	
Water Depth (m)			Meters To Drilled (m)	0	

Volume information (Salt Pol Mud)		
Surface System	800	bbl
Cased Hole Volume	890	bbl
New Hole Volume	0	bbl
Wash Out Volume	0	bbl
Sub Total	1,690	bbl
Excess Vol.	800	bbl
Sub Total	2,490	bbl
Carried from last section	0	bbl
Volume Received	0	bbl
Sub Total	2,490	bbl
New Vol Required	2,490	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Soda Ash	110	1-2	45
Caustic Soda	55	0.5 - 1	45
Salt (NaCl)	2,200	Up to 109	123
Starch LV	55	5 - 6	272
Limestone Powder	2,200	up to 114	129
Xanthan Gum	55	0.5 - 1	45
Biocide	55	As Requirement	

Contingency Materials			
Product Name	Function	Unit Size	QTY
Pipe Lax	Pipe freeing agent	55 Gal	2
De foamer	De foamer	55 Gal	2
Citric Acid	Alkalinity control	55 lb	20
Sodium Bicarbonate	CMT Contaminiation Remover	55 lb	20

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.

2-All the calculation based on max material Concentration which mentioned.

COMPLETION FLUID SECTION

The completion fluid for this section will be fresh water treated with a scavenger, inhibitor and biocide.

Solids-Free $CaCl_2$ Fluid Systems

This fluid is recommended to make from fresh water or is solutions made with water and the addition of one or more electrolytes. These fluids do not contain either suspended particles or viscosifiers, which mean neither filtration control nor viscosity control, are provided. Some Brine is corrosive and when corrosion inhibitors are required, compatibility studies should be performed. Some brine is toxic and precautions should be taken to prevent contact with unprotected skin. Brine solutions can be prepared to give good inhibition of clays or shales in producing formations. In addition, common contaminants do not affect the brines.

Salt solutions were formerly considered too corrosive to be used as work-over or completion fluids or as packer fluids. The corrosion problem can be solved by adjusting the pH of the solution to above 9 (min) to 10; however, pH measurements may be less important than measuring and controlling the p-alkalinity of the filtrate; this value should be kept above 1 cc. Lime be used for pH control.

Adjustment of pH, or alkalinity, does not necessarily ensure the solution to be corrosion resistant. It may be necessary to use a corrosion inhibitor to achieve lower corrosion rates. In some cases, an oxygen scavenger might be used in conjunction with a filming amine inhibitor.

Corrosion Control

Corrosion control in completion fluids is important to protect tubular goods and equipment. Corrosion can be limited by controlling the fluid pH at the neutral or alkaline level, and treating out dissolved oxygen. In severe environments such as high temperature or contamination with acid gases, the addition of filming amines can limit corrosion to acceptable levels.

The completion fluid for this section will be Local water treated with a scavenger, and inhibitor.

Pit management will be a key in the successful completion of this interval and sufficient time for planning and preparation should be allowed.

All valves should be checked for leakage beforehand and defects identified to be reported and rectified by drilling contractor.

The mud pits which are to be used to mix the completion brine, should emptied and cleaned adequately and associated lines flushed with water

The rig pumps are to displace all clean up pills as well as the $CaCl_2$ at intended weight. Prepare spacers and wash pills as per program in the tanks. Ensure that sufficient spacer is prepared to account for any dead tanks volume.

Prepare all inhibited $CaCl_2$ before starting the displacement process. After preparing the brine should be filtered through filtration unit and store in a clean pit before displacement.

Filtration Process:

Filtration provides clean fluids that will prevent damage to the formation, the reservoir and its permeability. It does this by removal of solids and particulates from the fluid that may otherwise block the open space within the rock formation containing the hydrocarbon fluid within the production zone.

Any solids and contaminates in the completion fluid will reduce its effectiveness. An efficiently filtered and completely solid free completion fluid will add value to both the productivity and the dependability of the well in the long term.

VIRA dual pod filtration cartridge unit is recommended for filtering brines and produced water in completion, stimulation, and workover operations. The pods are connected by a manifold, providing staged filtration capability. Staged filtration utilizes cartridge filters with different ratings to maximize filtration efficiency. Cartridge filtration is designed for use with fluids containing a low percent by volume of suspended solids. Cartridge filtration is not economic when fluids are contaminated with oil or other substances.

Total Solids can be measured at the wellsite using a turbidity meter. Solids that are not added to the system to enhance the performance of a brine are considered contamination. Contaminations include formation clays, precipitates, and polymer residues, among other things. These contaminations can be filtered at the well site using diatomaceous earth, a plate and frame press, and Five-micron to ten absolute cartridges.

Note: a clear completion fluid should not be sent to the wellsite with an NTU (Nephelometric Turbidity Unit) greater than 30-60 or a suspended-solid Concentration greater than 30-60 ppm and for fluid returning from the well NTU should be less than 150.

Displacement Process:

It is recommended to circulate and condition the mud prior to displacement and to spot the open hole section to conditioned mud or a solids-free fluid loss pill. Circulate and condition the mud as needed prior to pumping the clean out pills to optimum displacement properties).

After opening the circulating sub, the pumping sequence is as follows:

No.	Description	Volume BBL	Comment
1	Viscous pill	50	Clean pill
2	Continue pumping Completion Fluid until cleanliness specification has been met. Dump return at surface		

✓ **Viscous Spacer as clean pill**

NACL Brine or Fresh water
Xanthan Gum : Viscosifiers

- Mixing procedure:

1. Fill a tanks with required volume NACL Brine or Fresh water for pump 50 bbl Brine.
2. Add 3-4 lb/bbl Xanthan Gum.

✓ **Completion fluid**

$CaCl_2$ Brine at 85 pcf.
Raise PH to Above 9.5 with Lime
At the end and before pump to the well
Corrosion: Corrosion inhibitor Packer fluid design
H_2S Scavenger: to eliminate dissolved Hydrogen Sulfide from formation
O_2 Scavenger: Oxygen scavenger (to eliminate dissolved oxygen as source of corrosion)

Mixing procedure for 1 BBL 80 pcf Brine:

1. Fill the tank with 0.9 bbl. fresh water
2. Add 195 ppb $CaCl_2$.
3. Raise PH to above 9.5 with 3 - 4 ppb Lime
4. Add 1% V/V Corrosion Inhibitor
5. Add 0.5% V/V H_2S Scavenger
6. Add 0.5% V/V O_2 Scavenger

RIG CLEAN UP CHECK LIST (Should be done by Rig Operator)

A. Areas to clean.

1. Surface pits.
2. Clean the suction, return, settling, and reserve pits.
3. Clean the sand traps.
4. Clean the ditches and troughs.
5. Clean the trip tank.
6. Clean the shaker area including all solids equipment and lines.
7. Clean the mixing hoppers.
8. Clean the cement unit.
9. Clean any other pits or equipment that the completion fluid may contact.
10. Cycle all valves in mixing system.

B. Items to flush out.

1. All mud pumps and suction lines.
2. All centrifuges and lines.
3. Chemical and weighting mixing lines in contact with the completion fluid.
4. Lines from the pit room to the trip tank.
5. Lines from the rig floor to the trip tank.
6. Choke, choke manifold and kill lines.
7. Standpipes.
8. Degassing and solids control equipment.
9. Lines to the cement unit and/or other pumping equipment.
10. Casings fill up line.
11. Lines to and from the filtration line.
12. Take on and return lines to and from the boat.
13. Any other lines that the completion fluid may contact.

Displacement Procedure

Pre-Displacement Steps

1. Run bit and dual or tandem scrapers.
2. Circulate and condition mud. Reduce funnel viscosity as much as possible while maintaining adequate rheological properties to clean hole and prevent settling of solids. Maintain critical annular velocity to assure that liner tops are thoroughly cleaned prior to displacement.
3. Rig up equipment for displacement of completion fluid. All tanks, circulating lines, and pumps to be used in the displacement should be cleared of residual mud.
4. Insure sufficient space is available to store displaced mud.

DRILLING FLUID TYPE					
Mud Density (pcf)	85	-	85	LGS Below (%vol)	
Bit Size (inch)		0		Solid Control Equipment Eff. %	
Section TD (m)		3945		Hole Wash Out %	
Casing OD/ID (inch)				Dilution Factor (bbl/m)	
Water Depth (m)				Meters To Drilled (m)	

Volume information (NaCl Brine)		
Surface System	250	bbl
Cased Hole Volume	890	bbl
New Hole Volume	0	bbl
Wash Out Volume	0	bbl
Sub Total	1,140	bbl
Dilution Volume	0	bbl
Sub Total	1,140	bbl
30 % Extera Vol	342	bbl
Volume Received	0	bbl
Sub Total	1,482	bbl
New Vol Required	1,482	bbl
Vol transfer out	0	bbl

Product	Unit Size lb or bbl	Concentration	Expected Consumption
Lime	55	3-4	108
CaCl2	2,200	up to 195	112
Corrosion Inhibitor	55 Gal	1% V/V	10
H_2S Scavenger	55 Gal	0.5% V/V	5
O_2 Scavenger	55 Gal	0.5% V/V	5

1-Please note all calculation are based on normal hole condition and maximum SCE effieciency. Concentration and usage of material may be change as hole condition dictate.
2-All the calculation based on max material Concentration which mentioned.

Note: During preparing completion fluid, O2 scavenger should be added while all mixers were off.

7.1 Cement Program

Not applicable.

8. Completion

8.1 Proposed Completion string

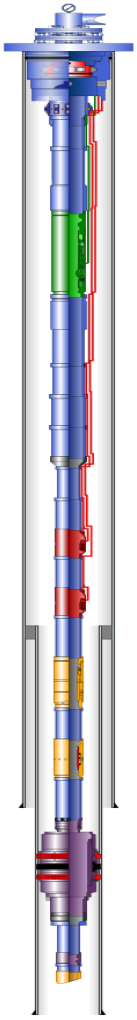
Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)				11.69	0.000
	2	TBG HANGER 11" * 4-1/2" ACME Box up * 4-1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type "H" Two Way	11	3.68	0.6	12.29	11.690
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* 9Cr1MO	5.5	3.64	1.01	13.300	12.290
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.5	60.800	13.300
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	62.420	60.800
	6	4-1/2" FLOW COUPLING,18.9#,9Cr1MO,HSM3 , 6 FT	5.5	3.64	1.72	64.140	62.420
	7	TRSSV Tubing Retrievable Saftey Valve, 4-1/2", 18.9# w/ 3.437" Seal bore,HSM3 ,9Cr1MO, 7.5K WP with AF Profile, Flapper Type	7.5	3.437	1.78	65.920	64.140
	8	4-1/2" FLOW COUPLING, 18.9#,9Cr1MO,HSM3 , 4' FT	5.5	3.64	1.11	67.030	65.920
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	68.650	67.030
	10	328 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	3116	3184.650	68.650
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.25	2.922	1.02	3185.670	3184.650
	12	18 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	171	3356.670	3185.670
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3358.490	3356.670
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3359.220	3358.490
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.8	3361.020	3359.220
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3370.520	3361.020
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3372.340	3370.520
	22	3-1/2" EXP. JOINT,HSM3 , 9Cr1MO, 10' STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 15,000 LBS,	5.315	2.922	5.755	3378.095	3372.340
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3379.917	3378.095
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3389.417	3379.917
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.625	3391.042	3389.417
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, 9Cr1MO, HSM3 with Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4.551	2.75	1.19	3392.232	3391.042
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.7	3393.932	3392.232
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.27	3403.202	3393.932
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.725	3404.927	3403.202
	31	7" 29 PPF CASING HYDRAULIC SET RETRIEVABLE PRODUCTION PACKER TO BE RUN IN 7.00" 29 PPF CASING	5.910	2.922	4.44	3409.367	3404.927
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3411.189	3409.367
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3420.689	3411.189
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.825	3422.514	3420.689
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, 9Cr1MO w/ Seal Bore 2.75" TYPE XN (NO-GO ID: 2.635")	4.5	2.75	0.37	3422.884	3422.514
	38	RE-ENTERYY MULE SHOE	4.25	2.922	0.37	3423.254	3422.884
		END OF COMPLETION STRING					3423.254
Tubing Retrievable Saftey Valve						65.920	64.140
Traveling Joint						3378.095	3372.340
SSD						3392.232	3391.042
7" HYDRAULIC SET RETRIEVABLE PRODUCTION PACKER						3409.367	3404.927
NO-GO Landing Nipple						3422.884	3422.514
Entry Guide						3423.254	3422.884

Figure 7: Proposed Completion Schematic

Note: Completion schematic and program will be issued in completion program in due time.

9. BOP, Wellhead & X-Mass Tree

9.1 Existing Wellhead & X-Mass Tree



Figure 8: Existing Wellhead

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Sarvak	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Sarvak & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Table 10: AZNS Wellhead Information

- **Wellhead Specification**

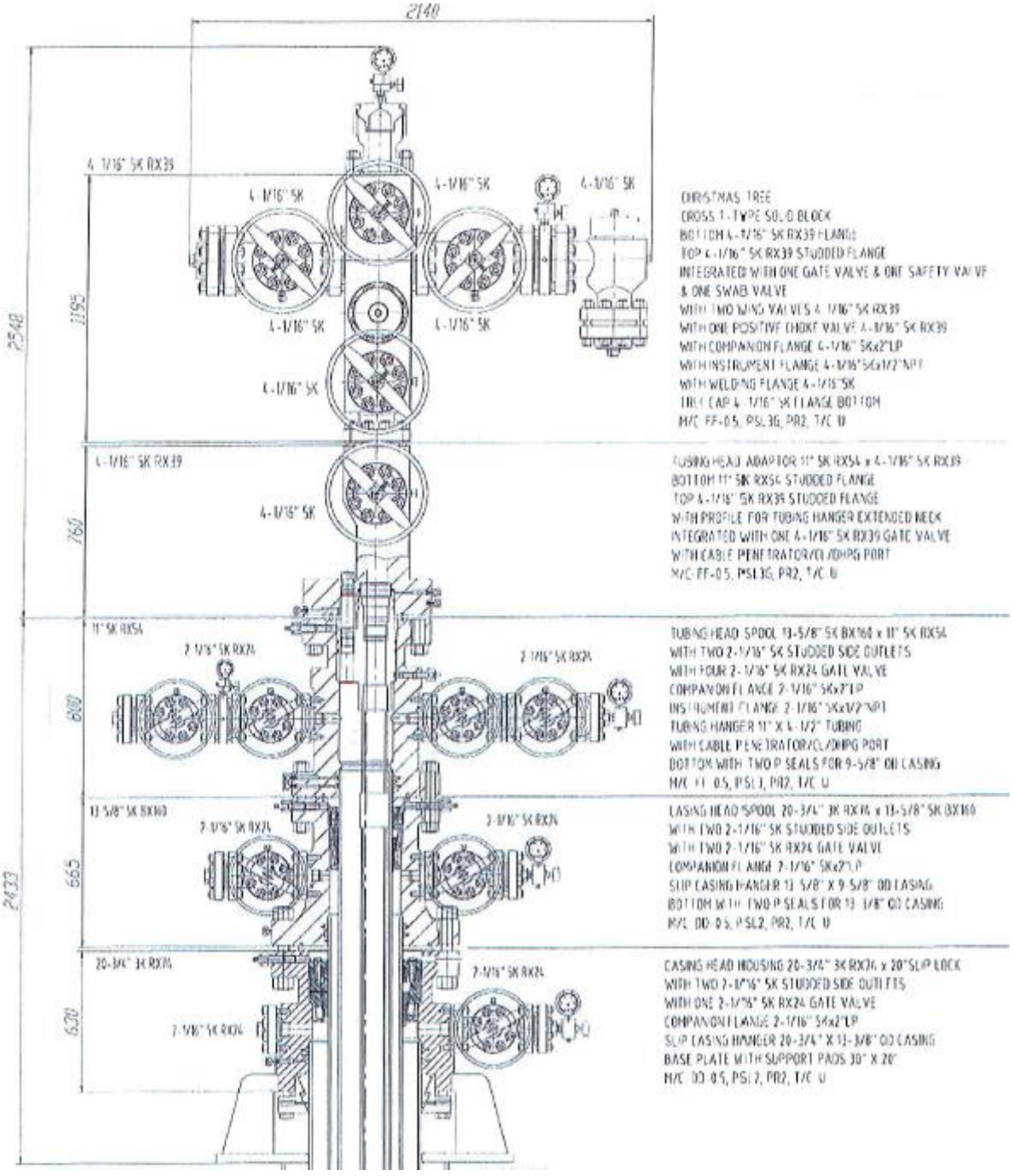
NOTE: Wellhead and X-Mass Tree already installed by client.

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm** Material of body: Forged low alloy steel 410SS

	G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Table 11: AZNS 122 Wellhead Specification

9.2 Proposed Wellhead schematic



9.3 BOP

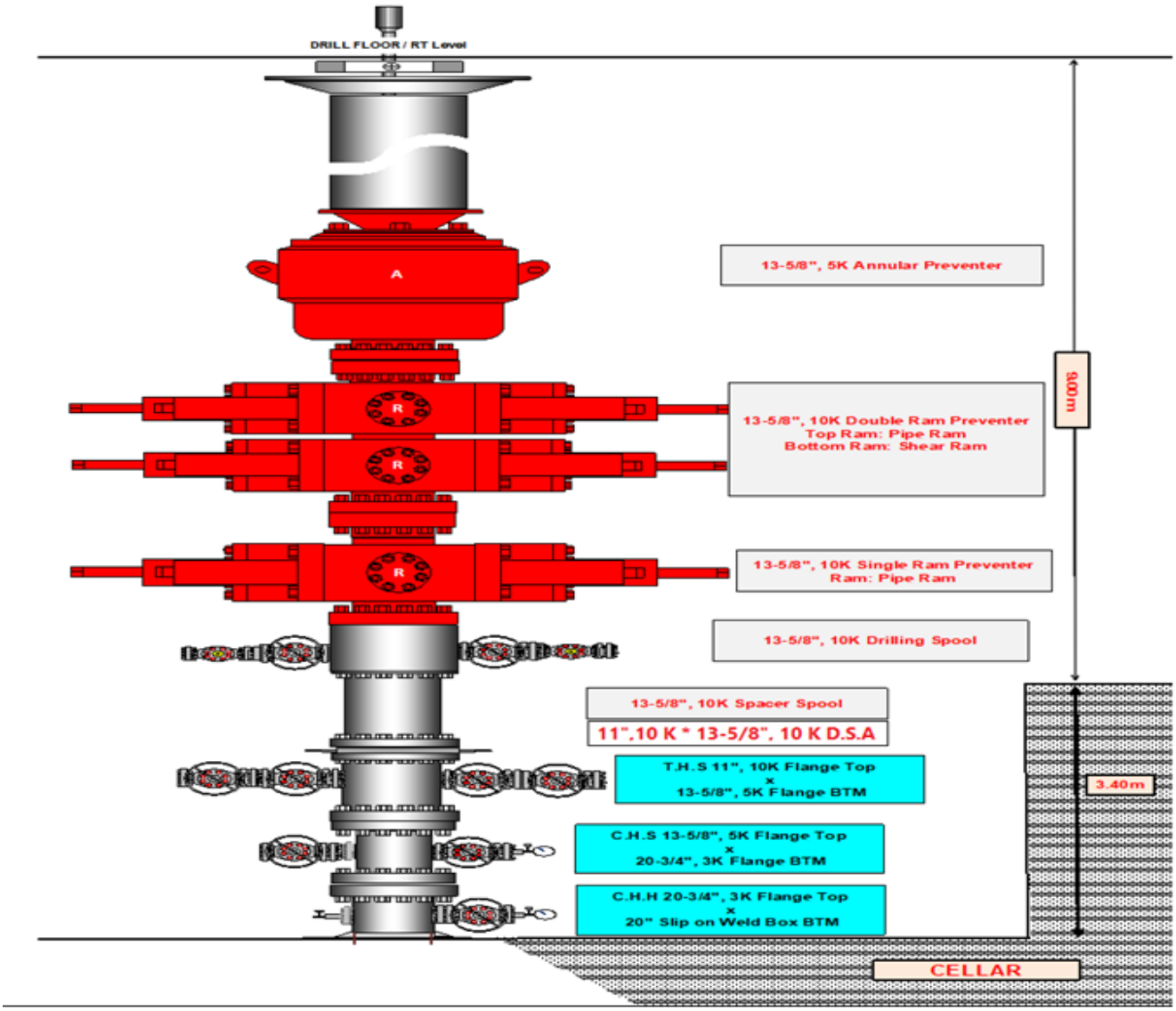


Figure 9: BOP Sketch

10.

10. Waste Management

In case of requirement to dispose wastewater in workover well, waste management program will be issued at due time as PEDEC policy.

11. Health, Safety and Environments

12.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA and contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

12.1.1 WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

12.1.2 Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

12.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System and well control equipment shall be recorded in the IADC daily reports and shall be signed by both the Drilling Contractor's Tool Pusher and Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

12.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.
While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

12.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.
Extreme care shall be taken to monitor mud volume, drilling breaks and gas cut mud.
The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.
The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.
The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.
The Driller is responsible for carefully measuring and recording the slow pump pressure.
Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure and record the slow pump pressure at three different flow rates with all pumps at the following times:
Once per tour.
Whenever changes occur in the mud density/rheology.
Whenever changes occur in the dimension and characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

12.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop and a static monitoring of the well shall be carried out. Space-out for all Pipe Rams and Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer and rams.

12.6 Well control during drilling and completion operation

The drilling design shall be strictly implemented by the drilling contractor and when it is discovered there is incompliance of downhole situation with design and the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, and safe operation can be affected, after the discussion between drilling rig manager and the drilling supervisor, practical technical measures can be taken first and then report to the top management. The Company shall approve and notice in time as per the drilling program amendment procedure.
The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program and curve of actual mud weight must be drawn during drilling operation. For exploratory well and key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone and shale shall be drawn by the comprehensive mud logging crew.
When entering oil and gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing and mud loss.
SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate and circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate and circulating pressure shall be recorded in Daily Drilling Report and Kill Sheet.
Once for every 12 hours in general;
After bit change or BHA change; 3) Mud weight change exceeds $^{3.75}$pcf.
Oil and gas invasion and overflow shall be checked with wiper trip and shall be conducted under the following situation:
Each time before POOH after drilling oil and gas formations.
Severe invasion of oil and gas occurs during drilling, but overflow does not occur before POOH.
POOH after killing well.
POOH after lowering mud weight in wellbore.
POOH after curing mud loss during drilling oil and gas formation.
POOH after drilling oil and gas formation and a prolonged period of time is needed for other operations (E logging, running casing, running tubing and DST etc.).
Regarding the shut-in methods selection
Soft shut-in method is recommended for gas well;
Hard shut-in method is recommended for oil well;
Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;
Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.

Flow check shall be conducted regularly via trip tank as follows

Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)

POOH to last casing shoe

Before POOH BHA through BOP stack

Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.

During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.

When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.

During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.

During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. if equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.

Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud. Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.

The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.

During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.

"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m^3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

12.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.

Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

12.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.

Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.

All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)

All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.

PSSR (Pre-Startup Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

12.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.

Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.

Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.

Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.

Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)

Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.

Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.

The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.

Local community are recommended.

Hazardous Wastes

Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site

Mud purifying circulation system.

Cooling water spraying and circulating system for drilling pump.

Special recovery tank for deposed oil.

Reserve house for valuable mud chemicals.

Automatic mud weighting supply system.

BOP stack is equipped according to drilling design.

Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location

Sewage pit Area of the sewage pit: 1500 m2.

Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.

Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.

Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.

Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.

Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.

After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.

Install fuel supply lines and keep them in good condition.

All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.

The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:

Paved, fenced, marked and illuminated

Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).

In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).
Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).
Capable enough to provide storage for the entire range of hazardous wastes generated during the work.
The area shall be provided with Spill Containment and means of preventing run-off to and from the area.
All waste containers to be stored shall be appropriately labelled.
Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

12. H_2S Monitoring & Prevention Measure

13.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

13.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

13. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.

14. Requirements

Item	Description	QTY.	Unit	Supplier
1	8-3/8" M.M.J Mill	2	Ea.	MSA
2	8 3/8” Tri Cone Bit	1	Ea.	MSA
3	6" or 6 1/8" M.M..J Mill	2	Ea.	MSA
4	6" or 6 1/8" Tri cone bit	1	Ea.	MSA
5	2"-4" Taper Tap	2	Ea.	MSA
6	4 1/8” M.M.J Mill	1	Ea..	MSA
7	4 1/2" * 3 1/2" Completion String 7.5K	1	set	MSA
8	Drift Gauge, API for 4 ½” 18.9 ppf Tubing	2	Ea.	MSA
9	Drift Gauge, API for 3 ½” 10.2 ppf Tubing	2	Ea.	MSA
10	Thread Seal compound, 10 kg/pail	10	Ea.	MSA
11	Thread Lock compound, API modified, each box contains 12 kit, 1 lb/kit	4	Ea.	MSA
12	Section C of Wellhead	1	Set	MSA

13	Section D of Wellhead	1	Set	MSA
14	7 1/16” 5K companion Flange	2	Ea.	PEDEC/PPL
15	X.O.S (7" VAM Pin x 4 1/2" IF Box)	1	Ea.	PEDEC/PPL
16	7" VAM Landing Joint suitable with TBG HGR 10" X 7"VAM X 9 5/8" NECK. (at least 12 m length)	1	Ea.	PEDEC/PPL
17	TBG Hanger BPV (Own way/Tow way) suitable for TBG HGR 10" X 7"VAM X 9 5/8" NECK with required redress kit. (Nominal size: 6 5/8")	1	Set	PEDEC/PPL
18	TBG Hanger BPV (Own way/Tow way) Setting/Retrieving Tool and 5K complete set Lubricator (Under Well Pressure) with required redress kit	1	Set	PEDEC/PPL
19	3SV Accessories (Protecton Sleeve, Seperation Sleeve, dummy valve suitable for 3SV BAKER model "BA", Landing nipple 5.950" with All Related Running/Pulling Tools) with required redress kit	1	Set	PEDEC/PPL
20	SSD Accessories (Shifting Tool, plug Equalizing Prong, etc. for BAKER MODEL "CMD", profile size" 3.125" With All Related Running/Pulling Tools) with required redress kit	1	Set	PEDEC/PPL
21	NO-GO L.NPL Accessories (Plug, Equalizing Prong, etc. With All Related Running/Pulling Tools) for 2.812" AF profile with required redress kit	1	Set	PEDEC/PPL
22	NO-GO L.NPL Accessories (Plug, Equalizing Prong, etc. With All Related Running/Pulling Tools) for 2.812" AF profile with required redress kit	1	Set	PEDEC/PPL
23	X.O.S 7" VAM Pin x 4 1/2" IF BOX	1	Set	PEDEC/PPL
24	7" VAM Pin Golpa. Flange (flange 7 1/16" 5K)	1	Set	PEDEC/PPL
25	9-5/8" & 7" Scraper	1+1	ea.	OEOC

26	5" DP,HWDP, 3 ½" DP & HWDP	as required		OEOC
27	Well Head Services	as required		Service Co.
28	Surface Well Test Services	as required		Service Co.
29	Completion Services	as required		Service Co.
30	Slick line Services	as required		Service Co.
31	Casing Running Services	as required		Service Co.
32	Wireline Logging	as required		Service Co.
33	Dry Test, DST, RTTS	as required		Service Co.
34	Cementing Services	as required		Service Co.
35	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
36	H2S Services	as required		Service Co.
37	Mud Logging Services	as required		Service Co.
38	Rig Services	as required		Service Co.

39	Drilling & Completion Fluid Services	as required		Service Co.

Note: Quantity & Spec to be conformed with contract's SOW.
Note: In case of any shortage or discrepancy, the contract's SOW will be reference.

Table 12: List of Requirements